



Yakeen Plus (2025)

Practice Test - 18

DURATION : 200 Minutes

DATE : 26/12/2024

M. MARKS : 720

Topics Covered

Physics :	Physics And Measurement, Kinematics, Laws Of Motion, Work, Energy, and Power, Rotational Motion
Chemistry :	Some Basic Concept of Chemistry, State of Matter, Solutions, Organic Chemistry: Some Basic principles and Techniques, (IUPAC Naming), Organic Chemistry: Some Basic principles and Techniques, (Isomerism), Organic Chemistry: Some Basic principles and Techniques, (GOC), Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure
Biology :	(Botany): Cell - The Unit of Life, Cell Cycle and Cell Division, The Living World, Biological Classification, Plant Kingdom (Zoology): Structural Organization in Animals, Animal Kingdom, Biomolecules, Breathing and Exchange of Gases

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are **four Section** in the Question Paper, **Section I, II, III & IV** consisting of Section-I (Physics from question number **1 to 50**), Section-II (Chemistry: from question number **51 to 100**), Section-III (Botany: from question number **101 to 150**) & Section IV (Zoology: from question number **151 to 200**), having 50 Questions in each Subject and each subject is divided in two Section, Section A consisting 35 questions (all questions are compulsory) and Section B consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

Practice Test-18

Yakeen Plus (2025)

Q1

SECTION - A (PHYSICS)

A ball is thrown vertically upwards from the ground and a student gazing out of the window sees it moving upward past him at 10 ms^{-1} . The window is at 15 m above the ground level. The velocity of ball 3 s after it was projected from the ground is: [Take $g = 10 \text{ ms}^{-2}$]

- (1) 10 m/s , up (2) 20 m/s , up
(3) 20 m/s , down (4) 10 m/s , down

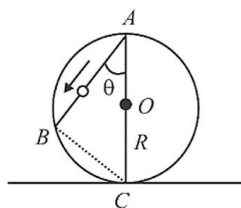
Q2 A particle is moving with speed $v = b\sqrt{x}$ along positive x -axis. Calculate the speed of the particle at time $t = \tau$ (assume that the particle is at origin at $t = 0$)

- (1) $\frac{b^2\tau}{4}$ (2) $\frac{b^2\tau}{2}$
(3) $b^2\tau$ (4) $4b\tau^2$

Q3 A body projected vertically upwards from the top of a tower reaches the ground in t_1 second. If it is projected vertically downwards from the same point with same speed, it reaches the ground in t_2 second. If it is just dropped from the top, it reaches the ground in t second. Find the relation between t , t_1 and t_2 .

- (1) $t = \sqrt{t_1 + t_2}$
(2) $t = t_1 t_2$
(3) $t = \sqrt{t_1 t_2}$
(4) $t = t_1 + t_2$

Q4 A frictionless wire AB is fixed on a sphere of radius R . A very small spherical ball slips on this wire. The time taken by this ball to slip from A to B is:



- (1) $\frac{2\sqrt{gR}}{g\cos\theta}$

- (2) $2\sqrt{gR} \cdot \frac{\cos\theta}{g}$
(3) $2\sqrt{\frac{R}{g}}$
(4) $\frac{gR}{\sqrt{g\cos\theta}}$

Q5 Given below are two statements:

Statement I: For one dimensional motion, the angle between acceleration and velocity may be zero.

Statement II: One dimensional motion is on a straight line.

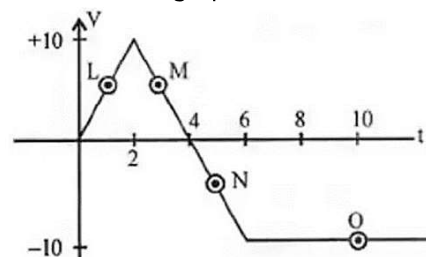
In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q6 The relation between time t and displacement x is $t = \alpha x^2 + \beta x$, where α and β are constants. The magnitude of retardation is (v is instantaneous speed)

- (1) $2\alpha v^3$
(2) $2\beta v^3$
(3) $2\alpha\beta v^3$
(4) $2\beta^2 v^3$

Q7 A particle starts from origin and moving along x -axis, whose v - t graph is as shown.



Choose the **Incorrect** statement.

- (1) At point L particle is speeding up.



- (2) At point M particle is moving in positive x -direction.
 (3) At point N particle is speeding up.
 (4) At point O particle is rest.

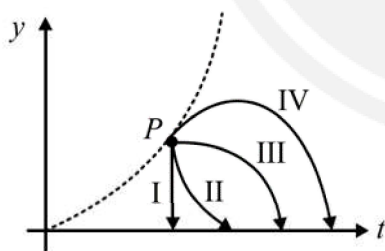
Q8 A projectile is given an initial velocity of $(\hat{i} + 2\hat{j})$ m/s, where, $\hat{i}$ is along the ground and $\hat{j}$ is along the vertical. If $g = 10 \text{ m/s}^2$, then the equation of its trajectory is

- (1) $y = x - 5x^2$ (2) $y = 2x - 5x^2$
 (3) $4y = 2x - 5x^2$ (4) $4y = 2x - 25x^2$

Q9 For a particle moving along $+x$ -axis, acceleration is given as $a = 2v^2$, where v is its instantaneous speed. If the speed of the particle is v_0 at $x = 0$, find speed as a function of x .

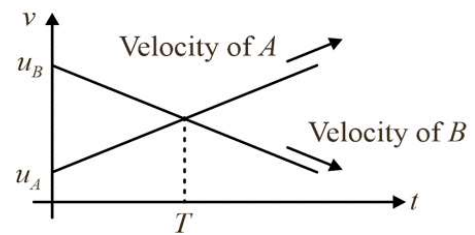
- (1) $v = v_0 e^{2x}$ (2) $v = v_0 e^x$
 (3) $v = v_0 \ln 2x$ (4) $v = v_0 \ln x$

Q10 An elevator is moving upwards with constant acceleration. The dashed curve in figure shows the position y of the ceiling of the elevator as a function of time t . At the instant indicated by the point P , a bolt breaks loose and drops from the ceiling. Which of the solid curves shown best describes the position of the bolt as a function of time?



- (1) I (2) II
 (3) III (4) IV

Q11 A situation is shown in which two objects A and B start their motion from same point in same direction. The graph of their velocities against time is drawn. u_A and u_B are the initial velocities of A and B respectively. T is the time at which their velocities become equal after start of motion. If the value of T is 4s, then the time after which A will meet B is:



- (1) 12s
 (2) 6s
 (3) 8s
 (4) data insufficient

Q12 A particle moves a distance x in time t according to equation $x = (t + 5)^{-1}$. The acceleration of particle is proportional to:

- (1) (velocity) $^{3/2}$
 (2) (distance) 2
 (3) (distance) $^{-2}$
 (4) (velocity) $^{2/3}$

Q13 A bomb is dropped from an aeroplane when it is at a height h directly above the target. If the aeroplane is moving horizontally with a speed v , the distance by which the bomb will miss the target is given by:

- (1) $2v\sqrt{\frac{h}{g}}$ (2) $v\sqrt{\frac{h}{g}}$
 (3) $v\sqrt{\frac{2h}{g}}$ (4) $v\sqrt{\frac{h}{2g}}$

Q14 A particle is projected at some angle with horizontal from the surface of the planet. The motion of the particle is described by the equation; $x = t$, $y = t - t^2$. Then match **List-I** with **List-II**:

List-I (quantity)		List-II (Magnitude only)	
(A)	velocity of projection	(I)	1
(B)	acceleration	(II)	$\sqrt{2}$
(C)	time of flight	(III)	2
(D)	maximum height attained	(IV)	$\frac{1}{4}$

Choose the **correct** answer from the options given below:

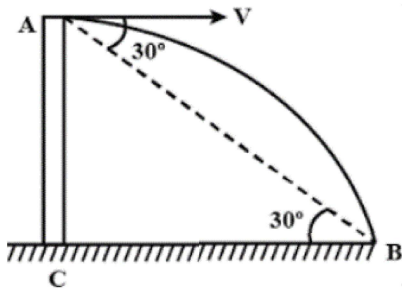
- (1) A-IV, B-I, C-II, D-III
 (2) A-IV, B-III, C-I, D-II



(3) A-II, B-III, C-I, D-IV

(4) A-III, B-IV, C-I, D-II

- Q15** An object is thrown horizontally from a point A from a tower and hits the ground 3s later at B. The line from A to B makes an angle of 30° with the horizontal. The initial velocity of the object is : (take $g = 10 \text{ m/s}^2$)



- (1) $15\sqrt{3} \text{ m/s}$ (2) 15 m/s
(3) $10\sqrt{3} \text{ m/s}$ (4) $25/\sqrt{3} \text{ m/s}$

- Q16** Two particles move in a uniform gravitational field with an acceleration g . At the initial moment the particles were located at one point and moved with velocities $u_1 = 3 \text{ ms}^{-1}$ and $u_2 = 4 \text{ ms}^{-1}$ horizontally in opposite directions. Find the distance (in m) between the particles when their velocity vectors become mutually perpendicular.
- (1) $7\frac{\sqrt{3}}{g}$ (2) $5\frac{\sqrt{3}}{g}$
(3) $14\frac{\sqrt{3}}{g}$ (4) $3\frac{\sqrt{3}}{g}$

- Q17** A particle starts in a 2D motion such that its position vector at any instant can be given as $\vec{r} = (2t^3 + 4t)\hat{i} + (3t^2 + 7)\hat{j}$. Find the velocity and acceleration of particle at $t = 1 \text{ sec}$, in m/s and m/s^2 respectively.
- (1) $\sqrt{180}, \sqrt{136}$ (2) $\sqrt{136}, \sqrt{180}$
(3) $\sqrt{140}, \sqrt{136}$ (4) $\sqrt{136}, \sqrt{140}$

- Q18** Find the percentage error in specific resistance given by $\rho = \frac{\pi r^2 R}{l}$, where r is the radius having value $(0.2 \pm 0.02) \text{ cm}$, R is the resistance of $(60 \pm 2) \text{ ohm}$ and l is the length of $(150 \pm 0.1) \text{ cm}$.
- (1) 23.4 (2) 12.2
(3) 11.1 (4) 7.8

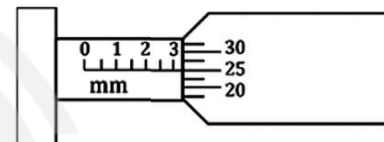
- Q19** The velocity v of a particle at time t is given by $v = at + \frac{b}{t+c}$, where a , b and c are constant.

The dimensions of a , b and c are respectively.

- (1) $[LT^{-2}]$, $[L]$ and $[T]$
(2) $[L^2]$, $[T]$ and $[LT^{-2}]$
(3) $[LT^2]$, $[LT]$ and $[L]$
(4) $[L]$, $[LT]$, and $[T^2]$

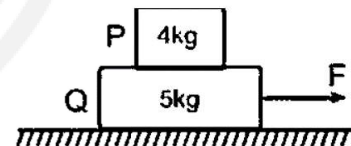
- Q20** The time dependence of physical quantity P is given by $P = P_0 e^{\alpha t^2 + \beta t + \gamma}$, where α , β , γ are constants and their dimensions are given by (where t is time):
- (1) $[M^0 L^0 T^{-2}]$, $[M^0 L^0 T^{-1}]$, $[M^0 L^0 T^0]$
(2) $[M^0 L^{-1} T^{-2}]$, $[M^0 L^0 T^{-1}]$, $[M^0 L^0 T]$
(3) $[M^0 L^0 T^{-1}]$, $[M L T^{-2}]$, $[M^0 L^0 T^{-1}]$
(4) $[MLT]$, $[M L T^0]$, $[M^0 L^0 T^0]$

- Q21** Find the reading of Screw gauge: [Least count = 0.01 mm]



- (1) 3.25 mm (2) 3.24 mm
(3) 3.30 mm (4) 3.41 mm

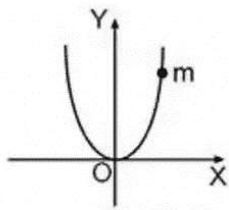
- Q22** In the given figure, the coefficient of friction between 4 kg and 5 kg blocks is 0.2 and between 5 kg block and ground is 0.1. Choose the **correct** statements.



- (1) Minimum force needed to cause system to move is 17 N.
(2) When force is 4 N static friction at all surfaces is 4 N to keep system at rest.
(3) Maximum acceleration of 4 kg block is 2 m/s^2 .
(4) Slipping between 4 kg and 5 kg blocks starts when $F > 17 \text{ N}$.

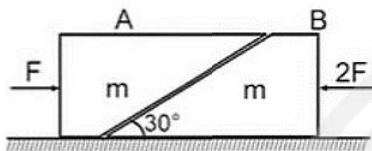
- Q23** A bead of mass m is located on a parabolic wire (equation $x^2 = ay$) with its axis vertical and vertex directed downward as in figure. If the coefficient of friction is μ , the highest distance above the x -axis at which the particle will be in equilibrium is:





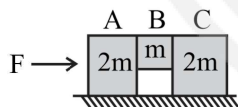
- (1) μa (2) $\mu^2 a$
 (3) $\frac{1}{4}\mu^2 a$ (4) $\frac{1}{2}\mu a$

- Q24** Two blocks 'A' and 'B' each of mass 'm' are placed on a smooth horizontal surface. Two horizontal forces F and $2F$ are applied on the two blocks 'A' and 'B' respectively as shown in figure. The block A does not slide on block B. Then the normal reaction acting between the two blocks is:



- (1) F (2) $F/2$
 (3) $\frac{F}{\sqrt{3}}$ (4) $3F$

- Q25** The system is pushed by a force F as shown in figure. All surfaces are smooth except between B and C. Friction coefficient between B and C is μ . Minimum value of F to prevent block B from slipping is:



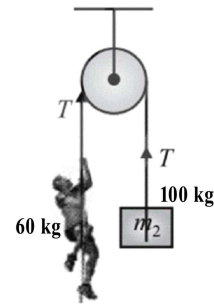
- (1) $\left(\frac{3}{2\mu}\right)mg$ (2) $\left(\frac{5}{2\mu}\right)mg$
 (3) $\left(\frac{5}{2}\right)\mu mg$ (4) $\left(\frac{3}{2}\right)\mu mg$

- Q26** A particle of mass 10^{-2} kg is moving along the positive x -axis under the influence of force $F(x) = -\frac{k}{2x^2}$, $k = 10^{-2} \text{ Nm}^2$. At $t = 0$, it is at $x = 1$ m and its velocity is $v = 0$. Find its velocity when it reaches $x = 0.5$ m.

- (1) $-1 \hat{i}$
 (2) $-2 \hat{i}$
 (3) $+1 \hat{i}$
 (4) $+2 \hat{i}$

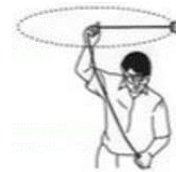
- Q27** In order to raise a mass of 100 kg, a man of mass 60 kg fastens a rope to it and passes the rope

over a smooth pulley. He climbs the rope with acceleration $5g/4$ relative to the rope. The tension in the rope is (take $g = 10 \text{ m/s}^2$)



- (1) $\frac{4875}{8} \text{ N}$ (2) $\frac{4875}{2} \text{ N}$
 (3) $\frac{4875}{4} \text{ N}$ (4) $\frac{4875}{6} \text{ N}$

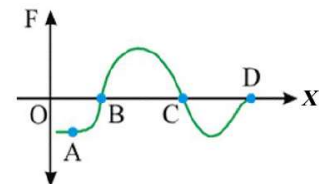
- Q28** When a stone is rotated with uniform speed in horizontal plane by means of a string, the magnitude of the momentum is fixed but its direction changes. A force is needed to cause this change in momentum vector.



This force is provided by:

- (1) gravity.
 (2) our hand through the string.
 (3) Both gravity and our hand through the string.
 (4) None of the above

- Q29** A particle is being acted upon by one dimensional conservative force. In the $F-x$ curve shown, four points A, B, C, D are marked on the curve. State which type of equilibrium does the particle have at position C?

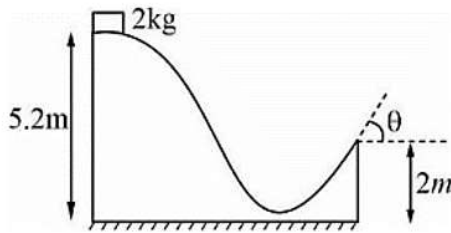


- (1) Stable equilibrium
 (2) Unstable
 (3) Neutral
 (4) No equilibrium

- Q30** Two identical blocks are simultaneously released from the tops of two smooth tracks that have identical shapes (as shown in figure) but different

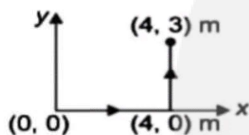


angles of projection. If $\theta = 30^\circ$ for first track and $\theta = 45^\circ$ for second track then difference between maximum heights attained by blocks in air will be: ($g = 10 \text{ ms}^{-2}$)



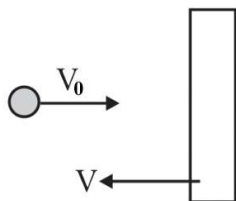
- (1) 0.6 m (2) 0.4 m
(3) 1 m (4) 0.8 m

- Q31** A force $\vec{F} = (3\vec{i} + 4\vec{j}) \text{ N}$ acts on a particle moving in $(x - y)$ plane. Starting from origin, the particle first goes along (x) -axis to the point $(4, 0) \text{ m}$ and then parallel to the (y) -axis to the point $(4, 3) \text{ m}$. The total work done by the force on the particle is:



- (1) +12 J (2) -6 J
(3) +24 J (4) -12 J

- Q32** A particle of mass m moves with velocity $v_0 = 20 \text{ m/sec}$ towards a large wall that is moving with velocity $v = 5 \text{ m/sec}$ towards the particle as shown. If the particle collides with the wall elastically, the speed of the particle just after the collision is:



- (1) 30 m/s (2) 20 m/s
(3) 25 m/s (4) 22 m/s

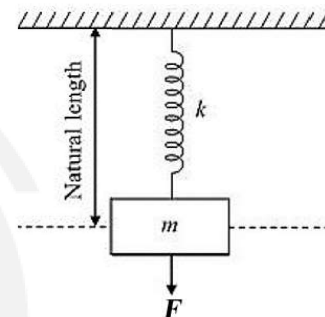
- Q33** The potential energy of a particle of mass 1 kg free to move along the x -axis is given by $U(x) = (3x^2 - 4x + 6) \text{ J}$. Force acting on the particle at $x = 0$ is:

- (1) $2\hat{i} \text{ N}$ (2) $-4\hat{i} \text{ N}$
(3) $5\hat{i} \text{ N}$ (4) $4\hat{i} \text{ N}$

- Q34** A particle of mass m moves in a circular path of constant radius r such that its centripetal acceleration a_c varies with time as $a_c = k^2 r t^2$ where k is a constant. The power delivered to the particle by the force acting on it is:

- (1) $2\pi m k^2 r^2 t$ (2) $m k^2 r^2 t$
(3) $\left(\frac{m k^4 r^2 t^5}{3}\right)$ (4) Zero

- Q35** Force $F = 3mg$ is applied on block initially at rest in vertical plane. Maximum elongation in the spring will be (initially spring was in its natural length)

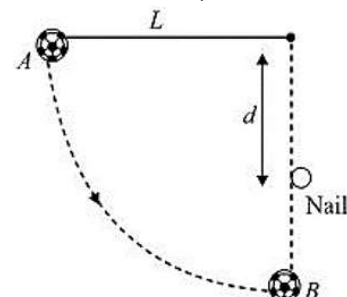


- (1) $\frac{8mg}{k}$ (2) $\frac{2mg}{k}$
(3) $\frac{4mg}{k}$ (4) $\frac{3mg}{k}$

Q36

SECTION - B (PHYSICS)

In the given system, when the ball of mass m is released from position A, it swings down the dotted arc along the vertical plane. A nail is located at a distance d below the point of suspension such that the ball just swings completely around a circle centered on the nail. Then minimum value of d possible is:



- (1) 0.6 L (2) 0.4 L
(3) 0.5 L (4) 0.3 L



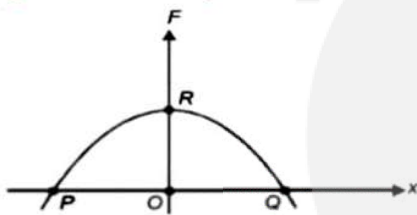
Q37 The potential energy of a 1 kg particle free to move along the X -axis is given by $U(x) = \left[\frac{x^4}{4} - \frac{x^2}{2} \right] J$. The total mechanical energy of the particle is 2J. Then, maximum speed (in ms^{-1}) is:

- (1) $\frac{3}{\sqrt{2}}$ (2) $\sqrt{2}$
 (3) $\frac{1}{\sqrt{2}}$ (4) 2

Q38 A spring of force constant k is cut into two pieces such that one piece is double the length of the other. Then, the long piece will have a force constant of

- (1) $(2/3)k$ (2) $(3/2)k$
 (3) $3k$ (4) $6k$

Q39 The variation of force F acting on a body moving along x -axis varies with its position (x) as shown in figure. The body is in stable equilibrium state at:

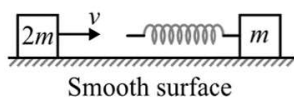


- (1) P (2) Q
 (3) R (4) Both P and Q

Q40 A body of mass 1 kg moving in the x -direction, suddenly explodes into two fragments of mass $1/8$ kg and $7/8$ kg. An instant later, the smaller fragment is 0.14 m above the x -axis. The position of the heavier fragment is:

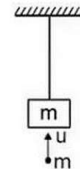
- (1) $\frac{1}{50}$ m above x -axis
 (2) $\frac{1}{50}$ m below x -axis
 (3) $\frac{7}{50}$ m below x -axis
 (4) $\frac{7}{50}$ m above x -axis

Q41 A block of mass m is attached with a spring of spring constant k as shown in the figure. Another block of mass $2m$ moving with speed v hits the spring. Speed of blocks at the instant of maximum compression of spring is:



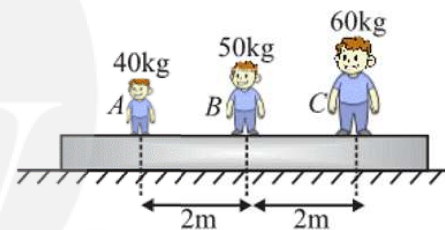
- (1) $\frac{v}{3}$ (2) $\frac{v}{2}$
 (3) $\frac{2v}{3}$ (4) $\frac{3v}{2}$

Q42 A bullet of mass m moving vertically upwards instantaneously with a velocity u hits the hanging block of mass m and gets embedded in it, as shown in the figure. The height through which the block rises after the collision. (assume sufficient space above block) is:



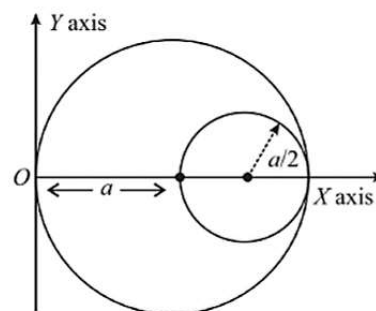
- (1) $u^2/2g$ (2) u^2/g
 (3) $u^2/8g$ (4) $u^2/4g$

Q43 Three men A, B and C of mass 40 kg, 50 kg and 60 kg are standing on a plank of mass 90 kg, which is kept on a smooth horizontal plane. If A and C exchange their positions then mass B will shift:



- (1) $1/3$ m toward left.
 (2) $1/3$ m towards right.
 (3) will not move w.r.t. ground.
 (4) $5/3$ m towards left.

Q44 A circular hole of radius $\left(\frac{a}{2}\right)$ is cut of a circular disc of radius ' a ' as shown in figure. The centre of mass of the remaining portion with respect to point ' O ' will be:



- (1) $\frac{5}{6}a$ (2) $\frac{1}{6}a$
 (3) $\frac{10}{11}a$ (4) $\frac{2}{3}a$

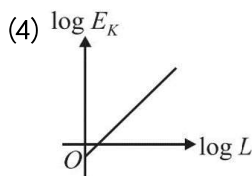
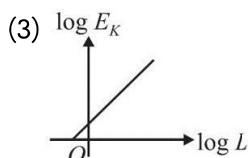
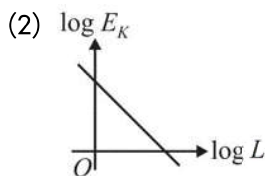
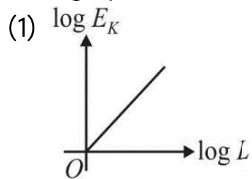


- Q45** A uniform rod smoothly pivoted at one of its ends is released from rest. If it swings in vertical plane, the maximum speed of the end P of the rod is:

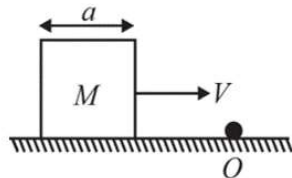


- (1) $2\sqrt{3gl}$ (2) $\sqrt{3gl}$
(3) $2\sqrt{2gl}$ (4) $\sqrt{gl}$

- Q46** If E_k is rotational K.E. and L is angular momentum, then graph between $\log L$ and $\log E_k$ will be:



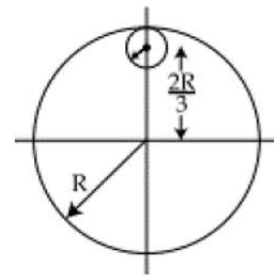
- Q47** A cubical block of side a is moving with velocity v on a horizontal smooth plane as shown. It hits a ridge at point O . The angular speed of the block after it hits O is:



- (1) $\frac{3v}{4a}$ (2) $\frac{3v}{2a}$
(3) $\frac{\sqrt{3}v}{\sqrt{2}a}$ (4) Zero

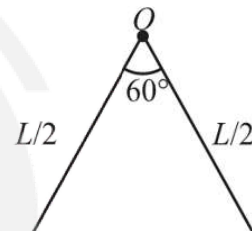
- Q48** From a uniform circular disc of radius R and mass $9M$, a small disc of radius $\frac{R}{3}$ is removed as shown in the figure. The moment of inertia of the remaining disc about an axis perpendicular to the

plane of the disc and passing through centre of disc is:



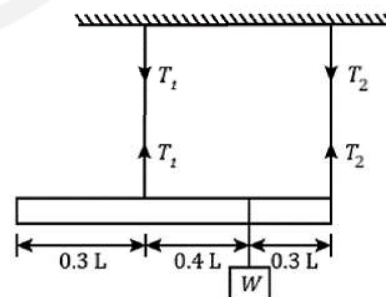
- (1) $\frac{37}{9}MR^2$ (2) $4MR^2$
(3) $\frac{40}{9}MR^2$ (4) $10MR^2$

- Q49** A thin rod of length L and mass M is bent at the middle point O at an angle of 60° . The moment of inertia of the rod about an axis passing through O and perpendicular to the plane of the rod will be:



- (1) $\frac{ML^2}{6}$ (2) $\frac{ML^2}{12}$
(3) $\frac{ML^2}{24}$ (4) $\frac{ML^2}{3}$

- Q50** In figure, the bar is uniform and weighing $500N$. How large must W be if T_1 and T_2 are to be equal?



- (1) $500N$ (2) $300N$
(3) $750N$ (4) $1500N$

Q51

SECTION - A (CHEMISTRY)

Which of the following has neither secondary nor tertiary hydrogen?

- (1) Isobutane (2) Isopentane
(3) Pentane (4) Neopentane



Q52 If electronegativity of A is 2.0 and that of B is 3.0, what is the percentage of covalent character of the bond A – B?

- (1) 19.5% (2) 80.5%
(3) 46% (4) 54%

Q53 Outer electronic configuration of X is $3d^5 4s^1$. It belongs to group number:

- (1) 16 (2) 15
(3) 5 (4) 6

Q54 The equivalent weight of a metal is 9 and vapour density of its chloride is 59.25. The atomic weight of the metal is:

- (1) 23.96 u (2) 27.37 u
(3) 36.38 u (4) 48.37 u

Q55 In which of the following pairs, both the species are **not** isostructural?

- (1) XeF_2 , I_3^-
(2) NH_3 , PH_3
(3) XeF_4 , XeO_4
(4) SiCl_4 , PCl_4^+

Q56 Identify the **incorrect** statement among the following:

- (1) Amongst isoelectronic species, smaller the positive charge on the cation, smaller is the ionic radius.
(2) Amongst isoelectronic species, greater the negative charge on the anion, larger is the ionic radius.
(3) Atomic radius of the elements increases as one moves down the first group of the periodic table.
(4) Atomic radius of the elements decreases as one moves across from left to right in the 2nd period of the periodic table.

Q57 A 20 mL sample of CuSO_4 solution was evaporated to dryness, leaving 0.967g of residue. What was the molarity of the original solution? (Molar mass of $\text{CuSO}_4 = 159.5 \text{ g/mol}$)

- (1) 48.4 M (2) 0.0207 M
(3) 0.0484 M (4) 0.303 M

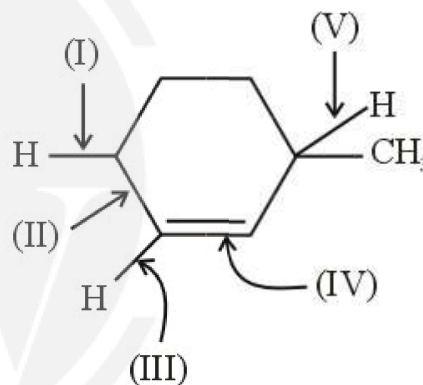
Q58 In the reaction;
 $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{l})$,
when 1 mol of ammonia and 1 mol of O_2 are made to react to completion:

- (1) 1.5 mol of H_2O is produced
(2) 1.5 mol of NO is produced
(3) all the O_2 is consumed
(4) all the NH_3 is consumed

Q59 Two elements A and B form two compounds AB_2 and AB_3 . If 0.25 mole of AB_2 weighs 16g and 0.25 mole of AB_3 weighs 20g, then the atomic mass of A and B are respectively is:

- (1) 16 and 32 g/mol
(2) 16 and 24 g/mol
(3) 32 and 16 g/mol
(4) 16 and 21 g/mol

Q60 Which of the following σ -bond participate in hyperconjugation?



- (1) (I) and (II) (2) (I) and (V)
(3) (II) and (V) (4) (III) and (IV)

Q61 Which of the following salt, will have the same value of van't Hoff factor (i) as that of $\text{K}_4[\text{Fe}(\text{CN})_6]$?

- (1) $\text{Fe}_2(\text{SO}_4)_3$
(2) NaNO_3
(3) $\text{Ca}(\text{NO}_3)_2$
(4) K_2SO_4

Q62 16 g of oxygen(O_2) and 3g of hydrogen(H_2) are mixed and kept at 760 mm of Hg pressure at 0°C . The total volume occupied by the mixture will be nearly:

- (1) 22.4 L (2) 33.6 L
(3) 448 L (4) 44800 mL



Q63 The **correct** decreasing priority order of the following groups in IUPAC naming of organic compounds is:

$-\text{OH}$, $-\text{C}\equiv\text{C}-$, $-\text{C}=\text{C}-$ and $-\text{NH}_2$ is:

- (1) $-\text{NH}_2 > -\text{C}=\text{C}- > -\text{C}\equiv\text{C}- > -\text{OH}$
- (2) $-\text{NH}_2 > -\text{C}=\text{C}- > -\text{OH} > -\text{C}\equiv\text{C}-$
- (3) $-\text{OH} > -\text{C}\equiv\text{C}- > -\text{C}=\text{C}- > -\text{NH}_2$
- (4) $-\text{OH} > -\text{NH}_2 > -\text{C}=\text{C}- > -\text{C}\equiv\text{C}-$

Q64 At 25°C , the total pressure of an ideal solution obtained by mixing 3 moles of A and 2 moles of B is 184 torr. What is the vapour pressure (in torr) of pure liquid B at the same temperature? (vapour pressure of pure liquid A at 25°C is 200 torr.)

- (1) 180 torr.
- (2) 160 torr.
- (3) 16 torr.
- (4) 100 torr.

Q65 Among the given sets, which represents the **correct** resonating structures?

- (1) $\text{H}-\text{C}\equiv\text{N}^+-\ddot{\text{O}}:-$ and $\text{H}-\ddot{\text{O}}:-\text{C}\equiv\text{N}:$
- (2) $\text{H}-\overset{+}{\text{O}}=\text{C}=\ddot{\text{N}}:-$ and $\text{H}-\ddot{\text{O}}:-\text{C}\equiv\text{N}:$
- (3) $\text{H}-\text{C}\equiv\text{N}^+-\ddot{\text{O}}:-$ and $\text{H}-\overset{\text{:O:}}{\parallel}{\text{C}}-\text{N}:$
- (4) $\text{H}-\ddot{\text{O}}:-\text{C}\equiv\text{N}^-$ and $\text{H}-\ddot{\text{N}}=\text{C}=\ddot{\text{O}}:$

Q66 Equal masses of SO_2 , CH_4 and O_2 are mixed in an empty container at 298 K, when total pressure is 2.1 atm. The partial pressure of CH_4 in the mixture is:

- (1) 0.5 atm
- (2) 0.75 atm
- (3) 1.2 atm
- (4) 0.6 atm

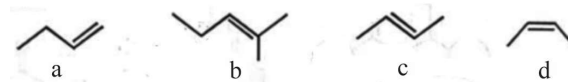
Q67 Europium is:

- (1) s-block element
- (2) p-block element

(3) d-block element

(4) f-block element

Q68 **Correct** decreasing order of heat of hydrogenation of the following alkenes is:



- (1) $b > c > d > a$
- (2) $b > d > c > a$
- (3) $a > c > d > b$
- (4) $a > d > c > b$

Q69 **Correct** decreasing order of dipole moment is:

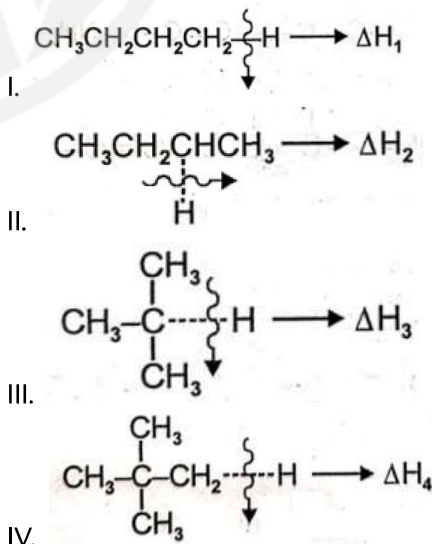
- (1) $\text{CH}_3\text{Cl} > \text{CH}_3\text{Br} > \text{CH}_3\text{F}$
- (2) $\text{CH}_3\text{Cl} > \text{CH}_3\text{F} > \text{CH}_3\text{Br}$
- (3) $\text{CH}_3\text{Br} > \text{CH}_3\text{Cl} > \text{CH}_3\text{F}$
- (4) $\text{CH}_3\text{Br} > \text{CH}_3\text{F} > \text{CH}_3\text{Cl}$

Q70 1.0 molal aqueous solution of an electrolyte A_2B_3 is 60% ionised. The boiling point of the solution at 1 atm is:

(K_b for H_2O is $0.52 \text{ K kg mol}^{-1}$)

- (1) 274.7 K
- (2) 377 K
- (3) 376.4 K
- (4) 374.7 K

Q71 The energy required to dissociate C-H bond of following compound for the formation of carbocation is given below,

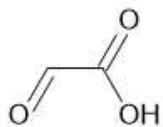


Minimum heat is associated with:

- (1) ΔH_1
- (2) ΔH_2
- (3) ΔH_3
- (4) ΔH_4



Q72 IUPAC name of the following compound is:



- (1) Glyoxalic acid
- (2) Formylmethanoic acid
- (3) 3-ketoethanoic acid
- (4) 2-ketoethanal

Q73 Which of the following is **incorrect** order of stability of carbocations?

- (1) $\text{Ph}_3\text{C}^+ > \text{Ph}_2\text{CH}^+ > \text{PhCH}_2^+$
- (2) $(\text{CH}_3)_3\text{C}^+ > (\text{CH}_3)_2\text{CH}^+ > \text{CH}_2=\text{CH}-\text{CH}_2^+$
- (3) $\text{CH}=\text{CH}-\text{CH}_2^+ > \text{CH}_3\text{CH}_2^+ > \text{HC}\equiv\text{C}^+$
- (4) $\left(\text{para}\right) \text{CH}_3\text{O}-\text{C}_6\text{H}_4\text{CH}_2^+ > \left(\text{para}\right) \text{CH}_3-\text{C}_6\text{H}_4\text{CH}_2^+ > \left(\text{para}\right) \text{NO}_2-\text{C}_6\text{H}_4\text{CH}_2^+$

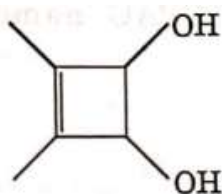
Q74 Which one of the following is **correct** order regarding the -I effect of the substituents?

- (1) $-\text{NR}_2 < -\text{OR} < -\text{F}$
- (2) $-\text{NR}_2 > -\text{OR} > -\text{F}$
- (3) $-\text{NR}_2 < -\text{OR} > -\text{F}$
- (4) $-\text{NR}_2 > -\text{OR} < -\text{F}$

Q75 Which one of the following has maximum number of nodal planes?

- (1) σ^*_{1s}
- (2) $\sigma^*_{2p_z}$
- (3) $\pi^*_{2p_x}$
- (4) $\pi^*_{2p_y}$

Q76 The IUPAC name of the compound is :



- (1) 1,2-dimethylcyclo-3,4-butene-diol
- (2) 1,2-dihydroxy-3,4-dimethylcyclobut-3-ene
- (3) 2,3-dimethylcyclobut-2-ene-1,4-diol
- (4) 3,4-dimethylcyclobut-3-ene-1,2-diol

Q77 Number of Cl^- and Ca^{2+} ions in 222 g of anhydrous CaCl_2 is:

- (1) $2N_A$ ions of Ca^{2+} and $2N_A$ ions of Cl^-
- (2) $2N_A$ ions of Cl^- and $4N_A$ ions of Ca^{2+}
- (3) $2N_A$ ions of Ca^{2+} and $4N_A$ ions of Cl^-
- (4) N_A ions of Ca^{2+} and $2N_A$ ions of Cl^-

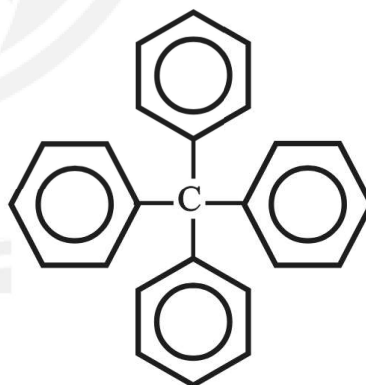
Q78 An aqueous solution of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) having a concentration of 34.2 g/L has an osmotic pressure of 2.38 atmospheres at 17°C . for an aqueous solution of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) to be isotonic with sucrose solution then concentration of glucose solution will be:

- (1) 34.2 g/L
- (2) 17.1 g/L
- (3) 18.0 g/L
- (4) 36.0 g/L

Q79 Ionic bond formation between A and B can take place only if :

- (1) ionisation enthalpy of A is low and electron affinity of B is high
- (2) ionisation enthalpy A is high and electron affinity of B is low
- (3) ionisation enthalpy A is high and electron affinity of B is high
- (4) ionisation enthalpy A is low and electron affinity of B is low

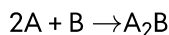
Q80 The **correct** IUPAC name of the compound is:



- (1) Tetraphenylmethane
- (2) neophenylethane
- (3) 3,3-diphenyldodecane
- (4) Tetrabenzene

Q81 Two substances A and B were made to react to form a third substance A_2B ; According to the following reaction,





Which of the following statements concerning this reaction is/are **incorrect**?

- (i) The product A_2B shows the properties of substances A and B.
- (ii) The product will always have a fixed composition.
- (iii) The product so formed cannot be classified as a compound.
- (iv) The product so formed is an element.

- (1) (i), (ii) and (iii)
- (2) (ii), (iii) and (iv)
- (3) (i), (iii) and (iv)
- (4) (iii) and (iv)

Q82 The ionic radii (in Å) of N^{3-} , O^{2-} and F^- are respectively:

- (1) 1.71, 1.40 and 1.36 Å
- (2) 1.71, 1.36 and 1.40 Å
- (3) 1.36, 1.40 and 1.71 Å
- (4) 1.36, 1.71 and 1.40 Å

Q83 Which of the following compound has maximum degree of unsaturation?

- (1)
- (2)
- (3)
- (4)

Q84 Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A) : XeF_2 is linear but H_2O is angular though both are AB_2 type molecules.

Reason (R) : F is more electronegative than H.

In the light of the above statements, choose the **correct** answer from the options given below:

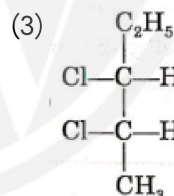
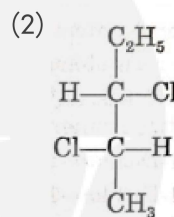
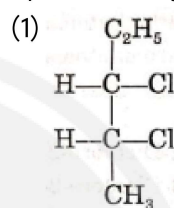
- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

Q85 In a regular octahedral molecule MX_6 , the number of $X - M - X$ bonds at 180° is:

- (1) three
- (2) two
- (3) six
- (4) four

Q86 **SECTION - B (CHEMISTRY)**

Which of the following compounds can exhibit optical activity?



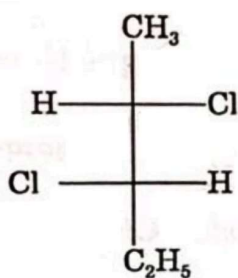
- (4) All of these.

Q87 One kilogram of a sea water sample contains 6 mg of dissolved O_2 . The concentration of O_2 in ppm is:

- (1) 10
- (2) 6
- (3) 10^6
- (4) 6×10^6

Q88 The absolute configuration of the following compound is:





- (1) 2S, 3R (2) 2S, 3S
(3) 2R, 3S (4) 2R, 3R

Q89 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion : 1 mole of $\text{H}_2(\text{g})$ and $\text{O}_2(\text{g})$ each gas occupies 22.4 L at standard temperature and pressure.

Reason : Molar volume for all gases at the standard temperature and pressure has the same value.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

Q90 Which of the following compounds do **not** exhibit tautomerism?

- (1) Acetone
(2) Benzophenone
(3) Benzaldehyde
(4) Both (2) and (3)

Q91 The density of a liquid is 1.2 g/mL. There are 35 drops in 2 mL. The number of molecules in one drop of liquid is:

(Molar mass of liquid = 70 g/mol)

- (1) $\left(\frac{1.2}{35}\right) N_A$ (2) $\left(\frac{1}{35}\right)^2 N_A$
(3) $\frac{1.2}{(35)^2} N_A$ (4) $1.2 N_A$

Q92 **Correct** order of decreasing X-O bond length is: (where X is the central atom)

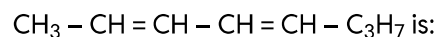
- (1) $\text{ClO}_4^- > \text{SO}_4^{2-} > \text{PO}_4^{3-} > \text{SiO}_4^{4-}$
(2) $\text{SiO}_4^{4-} > \text{PO}_4^{3-} > \text{SO}_4^{2-} > \text{ClO}_4^-$

- (3) $\text{SiO}_4^{4-} > \text{PO}_4^{3-} > \text{ClO}_4^- > \text{SO}_4^{2-}$
(4) $\text{SiO}_4^{4-} > \text{SO}_4^{2-} > \text{PO}_4^{3-} > \text{ClO}_4^-$

Q93 Which one of the following pairs represents stereoisomerism?

- (1) Chain isomerism and rotational isomerism
(2) Structural isomerism and geometric isomerism
(3) Linkage isomerism and geometric isomerism
(4) Optical isomerism and geometric isomerism

Q94 The number of geometrical isomers in case of a compound with the structure

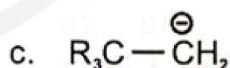
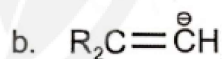
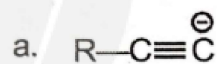


- (1) 4 (2) 3
(3) 5 (4) 2

Q95 Which of the following group of elements have the same number of electrons in their outermost shell?

- (1) Elements with atomic numbers 30, 48, 80
(2) Elements with atomic numbers 14, 15, 16
(3) Elements with atomic numbers 20, 30, 50
(4) Elements with atomic numbers 10, 18, 26

Q96 The stability of carbanions in the following



is in the order of:

- (1) $a > c > b$
(2) $a > b > c$
(3) $c > b > a$
(4) $b > c > a$

Q97 The geometrical arrangement of electron pairs and shape of I_3^- are respectively:

- (1) trigonal bipyramidal geometry, linear shape
(2) hexagonal geometry, T-shape
(3) triangular planar geometry, triangular planar shape
(4) tetrahedral geometry, pyramidal shape.



- Q98** Among LiCl , BeCl_2 , BCl_3 and CCl_4 , the covalent bond character varies as:
- (1) $\text{LiCl} < \text{BeCl}_2 > \text{BCl}_3 > \text{CCl}_4$
 - (2) $\text{BCl}_3 > \text{BeCl}_2 > \text{CCl}_4 > \text{LiCl}$
 - (3) $\text{LiCl} < \text{BeCl}_2 < \text{BCl}_3 < \text{CCl}_4$
 - (4) $\text{LiCl} > \text{BeCl}_2 > \text{BCl}_3 > \text{CCl}_4$
- Q99** Molality of 95% by volume of H_2SO_4 solution, whose specific gravity is 1.98 g mL^{-1} is:
- (1) 7.412 m
 - (2) 8.412 m
 - (3) 9.412 m
 - (4) 10.412 m
- Q100** The bond angles of NH_3 , NH_4^+ and NH_2^- are in the order of:
- (1) $\text{NH}_2^- > \text{NH}_3 > \text{NH}_4^+$
 - (2) $\text{NH}_4^+ > \text{NH}_3 > \text{NH}_2^-$
 - (3) $\text{NH}_3 > \text{NH}_2^- > \text{NH}_4^+$
 - (4) $\text{NH}_3 > \text{NH}_4^+ > \text{NH}_2^-$

Q101 **SECTION - A (BOTANY)**

Which of the following help attach the bacteria to the host tissues?

- (1) Cell wall and capsule
- (2) Mesosome and glycocalyx
- (3) Plasma membrane, cell wall and slime layer
- (4) Fimbriae

- Q102** Which of the following options represents the decreasing order of common characters among the taxonomic categories?
- (1) Family $\rightarrow$ Genus $\rightarrow$ Species
 - (2) Family $\rightarrow$ Order $\rightarrow$ Class
 - (3) Kingdom $\rightarrow$ Phylum $\rightarrow$ Class
 - (4) Division $\rightarrow$ Order $\rightarrow$ Genus

- Q103** Select the **correct** statement regarding euglenoids.
- (1) The cell wall of euglenoids have stiff cellulose plates on the outer surface.
 - (2) They have two flagella.
 - (3) One flagellum lies longitudinally and the other transversely in a furrow between the wall plates.
 - (4) Pigments of euglenoids are identical to those present in fungi.

- Q104** Read the following statements and mark the **correct** option.
- A. Interphase represents the phase between two successive M phases.
 - B. M phase represents the phase when actual cell division occurs.
 - C. M phase constitutes more than 95% of the duration of cell cycle
 - D. Interphase represents the cytokinesis.
- Choose the most appropriate answer from the options given below:
- (1) A and C only
 - (2) B and D only
 - (3) A and B only
 - (4) A, B and C only

- Q105** A prokaryotic autotrophic nitrogen fixing symbiont is found in:
- (1) *Cycas*.
 - (2) *Laminaria*.
 - (3) *Pinus*.
 - (4) *Selaginella*.

- Q106** Select the **mismatched** pair.

(1)	Chromosomes move to equator	Metaphase
(2)	Centromere splits	Anaphase
(3)	Aster formation	Prophase
(4)	Best stage to study morphology of chromosome	Anaphase

- Q107** Select the **incorrect** statements.
- A. Diplotene is characterised by the presence of chiasmata.
 - B. In spermatocytes of some vertebrates, diplotene can last for months or years.
 - C. Interkinesis is generally short lived.
 - D. The centriole duplicates during the S-phase of cell cycle in nucleus.
- Choose the most appropriate answer from the options given below:
- (1) A and C only
 - (2) B and D only



- (3) A and D only
(4) A, B and C only

Q108 Read the following statements and select the **incorrect** for fungi.

- (1) *Saccharomyces* is multicellular.
(2) The network of hyphae is known as mycelium.
(3) The cell walls of fungi are composed of chitin and polysaccharides.
(4) Coenocytic hyphae is characteristic of *Albugo*.

Q109 Zoospore is absent in _____ algae.

- (1) brown (2) green
(3) red (4) All of these

Q110 Match **List-I** with **List-II**:

List-I		List-II	
(A)	Diptera	(I)	Order
(B)	Insecta	(II)	Class
(C)	Arthropoda	(III)	Phylum
(D)	Muscidae	(IV)	Family

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-II, D-I
(4) A-I, B-II, C-III, D-IV

Q111 Identify from the following which is/are related to eukaryotic cell.

- A. Plant cell wall contain hemicellulose.
B. It has 70S type of ribosome present in the thylakoids.
C. Microbodies are present in plants only.
D. Centrioles have 9+2 arrangement of microtubules.

Choose the most appropriate answer from the options given below:

- (1) A, B, C and D are correct
(2) A, B and C are incorrect only
(3) A, B and C are correct only
(4) B, C and D are incorrect only

Q112 Which of the following will result in haploid cell formation?

- (1) Cell division by mitosis in diploid cell.
(2) Cell division by meiosis in haploid cell.
(3) Cell division by mitosis in haploid cell.

- (4) Both (2) and (3)

Q113 Given below are two statements.

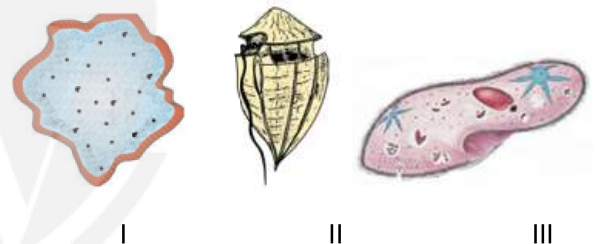
Statement I: The minor lipids are phospholipids that are arranged in a bilayer in plasma membrane.

Statement II: In addition to proteins, membrane also contains cholesterol.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct and Statement-II is incorrect.
(2) Statement-I is incorrect and Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

Q114 In following three diagrams identify **I**, **II** and **III** respectively and find out the **correct** match among the following set of options



	I	II	III
(1)	Spores possess true walls	Stiff cellulose plates	Two flagella
(2)	Stiff cellulose plates	Spores possess true walls	Spores possess true walls
(3)	Spores possess true walls	Stiff cellulose plates	Gullet
(4)	Stiff cellulose plates	Cilia	Spores possess true walls

Q115 Which of the following was **not** the feature of natural classification systems?

- (1) Based on embryology
(2) Based on phytochemistry.



- (3) Given by George Bentham and Joseph Dalton Hooker.
 (4) Based on mainly the androecium structure

Q116 Given below are two statements:

Statement I: Chemosynthetic autotrophic bacteria reduce various inorganic substances such as nitrates.

Statement II: Chemosynthetic autotrophic bacteria deplete nutrients like nitrogen, phosphorous, iron and sulphur.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

Q117 Arrange the following events of meiosis in the **correct** order from initial stage to later stage

- A. Terminalisation
 B. Crossing over
 C. Synapsis
 D. Chromatin condensation

Choose the most appropriate answer from the options given below:

- (1) $D \rightarrow C \rightarrow B \rightarrow A$
 (2) $C \rightarrow B \rightarrow A \rightarrow D$
 (3) $B \rightarrow A \rightarrow D \rightarrow C$
 (4) $A \rightarrow D \rightarrow C \rightarrow B$

Q118 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The most notable disease caused by viroids is bovine spongiform encephalopathy (BSE).

Reason R: Viroids are found to be a free RNA.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.

- (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.

Q119 Which of the following statement is **incorrect** with respect to lysosomes?

- (1) Lysosomes are simple tiny spherical sac-like structures.
 (2) They are distributed in the cytoplasm of the cell.
 (3) The enzymes of lysosomes work in basic condition.
 (4) The enzymes of lysosomes are synthesised by rough endoplasmic reticulum.

Q120 Given below are two statements.

Statement I: External and internal structure are considered in modern taxonomic studies.

Statement II: Diptera includes Felidae and Canidae.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct and Statement II is incorrect.
 (2) Statement I is incorrect and Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

Q121 Match List-I with List-II

List-I			List-II
(A)	Plantae	(I)	Loose tissue organisation
(B)	Monera	(II)	Nuclear membrane absent
(C)	Fungi	(III)	Organ and tissue level of organisation
(D)	Protista	(IV)	Cellular and eukaryotic



Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
 (2) A-II, B-IV, C-III, D-I
 (3) A-III, B-II, C-I, D-IV
 (4) A-I, B-II, C-III, D-IV

Q122 Read the following statements:

- A. Each different kind of plant, animal or organism, represents a species.
 B. The number of species that are known and described range between 17-18 million.
 C. Local names would vary from place to place, even within a country.
 D. If we visit a dense forest, we would probably see a much greater number and less kinds of living organisms in it.

Choose the most appropriate answer from the options given below:

- (1) B and C are correct only
 (2) A, B and C are incorrect only
 (3) A, B and C are correct only
 (4) B and D are incorrect only

Q123 Cellular organelles with membranes are:

- (1) lysosomes, Golgi apparatus and mitochondria.
 (2) nuclei, ribosomes and mitochondria.
 (3) Chromosomes, ribosomes and endoplasmic reticulum
 (4) Endoplasmic reticulum, ribosomes and nuclei

Q124 In bryophytes, male sex organ is:

- (1) unicellular antheridium.
 (2) multicellular antheridium.
 (3) multicellular antheridium and multiflagellate.
 (4) unicellular spermatogonium and non-motile.

Q125 Which of the following is **wrongly** matched in the given table?

Taxonomic category	Taxa
(1) Family	Poales
(2) Kingdom	Plantae
(3) Genus	<i>Triticum</i>
(4) Species	<i>aestivum</i>

Q126 Select the **incorrect** statements.

- A. Meiosis involves pairing of homologous chromosomes and recombination between them.
 B. Recombination nodule is formed at the end of meiosis-II.
 C. Meiosis involves four cell division.
 D. Meiosis –II is initiated after the parental chromosome replication.

Choose the most appropriate answer from the options given below:

- (1) A, B and C only
 (2) B, C and D only
 (3) A, C and D only
 (4) B and C only

Q127 Aplanospores are:

- A. motile spores
 B. non-motile spores
 C. sexual spores
 D. asexual spores
 E. exogenously produced.

Choose the most appropriate answer from the options given below:

- (1) (A),(C) and (E) only
 (2) (A),(D) and (E) only
 (3) (B),(C) and (E) only
 (4) (B) and (D) only

Q128 Select the **incorrect** statements.

- A. Dyad of cells is formed during the telophase-II stage of meiosis
 B. Interkinesis is stage during mitosis.
 C. Bivalent formation occurs during cytokinesis.
 D. Significance of meiosis lies in maintaining consistency of chromosome number during sexual reproduction.

Choose the most appropriate answer from the options given below:

- (1) A, B and C only
 (2) A, B and D only
 (3) A and C only
 (4) C only

Q129 Consider the following statements.

- A. All protozoans are heterotrophs.



- B. Protozoans are believed to be primitive relatives of animals.
 C. False feet is the characteristic of Flagellated protozoan.
 D. Protozoan belong to slime moulds.

Choose the most appropriate answer from the options given below:

- (1) A and B are correct only
 (2) A and B are incorrect only
 (3) A, B and C are correct only
 (4) A, B, C and D are incorrect

Q130 Green algae used as a food by space travellers:

- (1) *Ulothrix*. (2) *Porphyra*.
 (3) *Chlorella*. (4) *Chlamydomonas*.

Q131 Chlorophyceyan members are usually A due to the dominance of pigments B.

	A	B
(1)	olive green	chlorophyll <i>a</i> and carotene
(2)	red	chlorophyll <i>a</i> and phycoerythrin
(3)	dark green	chlorophyll <i>a</i> , <i>c</i> and carotenoids
(4)	grass green	chlorophyll <i>a</i> and <i>b</i>

Q132 Select the **correct** option that are present in protist.

- (1) Inclusion bodies and cell wall
 (2) Plasmid and plasma membrane
 (3) Mesosome and nucleoid
 (4) Plasma membrane and nucleus

Q133 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Large central vacuole is present in all eukaryotic cells.

Reason R: Membrane of vacuole is tonoplast.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.

- (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.

Q134 Select the **odd** one w.r.t taxa of Mango.

- (1) Sapindales (2) Poaceae
 (3) Dicotyledonae (4) Angiospermae

Q135 How many of the following are **correct** for the importance of the following plant.

Feature	Plant
Algin	Brown algae
Carrageen	Red algae
Agar	<i>Gelidium</i>
Packing material	<i>Sphagnum</i>
Prevent soil erosion	Mosses

Choose the most appropriate answer from the options given below:

- (1) Two (2) Four
 (3) Five (4) One

Q136

SECTION - B (BOTANY)

Select **correct** for basidiomycetes .

- (1) It is cyanobacteria.
 (2) It fixes nitrogen in paddy fields.
 (3) Karyogamy take place in the basidium.
 (4) The sexual spores are generally not found.

Q137 Water moves across the membrane by P while Na^+/K^+ move across plasma membrane by Q. Identify **P** and **Q** respectively.

P	Q
(1) passive transport	active transport
(2) osmosis	active transport



(3)	active transport	passive transport
(4)	active transport	active transport

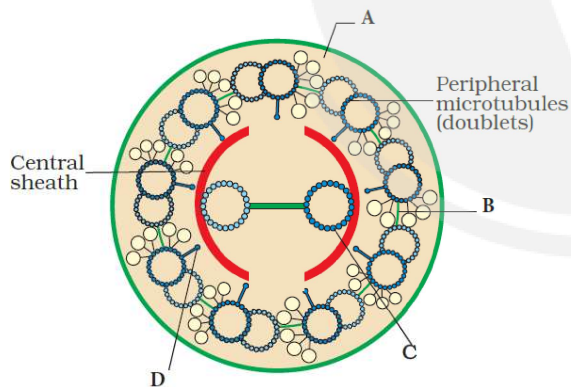
Q138 Match List-I with List-II:

List-I		List-II	
(A)	Thylakoids	(I)	Condensed structure of DNA
(B)	Cristae	(II)	Flat membranous sacs in stroma
(C)	Cisternae	(III)	Infoldings in mitochondria
(D)	Chromatin	(IV)	Disc-shaped sacs in Golgi apparatus

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
 (2) A-IV, B-III, C-II, D-I
 (3) A-III, B-IV, C-II, D-I
 (4) A-II, B-I, C-III, D-IV

Q139 Go through the section of cilia / flagella showing the different parts



In which of the following options correct words for all the four labellings A, B, C, and D are indicated?

- (1) A - Plasma membrane, B - Interdoublet bridge, C - Central microtubule, D-Radial spoke
 (2) A-Plasma membrane, B - Arm, C-Central microtubule, D-Radial spoke
 (3) A - Plasma membrane, B - Interdoublet bridge, C - Hub, D - Radial spoke
 (4) A - Plasma membrane, B - Interdoublet bridge, C - Hub, D-Arm

Q140 Select the option in which the scientific name is printed **correctly**.

- (1) *Solanum nigrum*
 (2) *Panthera Leo*
 (3) *Mangifera indica*
 (4) *Homo sapiens*

Q141 Multicellular rhizoids are the feature of:

- (1) *Ulothrix*. (2) *Ectocarpus*.
 (3) *Funaria*. (4) *Fucus*.

Q142 In class phaeophyceae, the plant body is usually attached to the substratum by a**A**.... and has a stalk, the**B**.... and leaf like photosynthetic organ-the....**C**.....Identify **A**, **B** and **C** respectively.

- (1) A – frond, B – stipe, C – holdfast
 (2) A – stipe, B – holdfast, C – frond
 (3) A – holdfast, B – stipe, C – frond
 (4) A – stipe, B – frond, C – holdfast

Q143 Identify from the following which is **not** a feature of chrysophytes.

- (1) In diatoms the cell walls form two thin overlapping shells.
 (2) Diatomaceous earth is gritty.
 (3) Golden algae are the chief ‘producers’ in the oceans.
 (4) Most of them are photosynthetic.

Q144 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Thallus of bryophytes is attach to substratum by unicellular and multicellular rhizoids.

Reason R: Thallus of bryophytes is erect or prostrate.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.



Q145 How many of the following are **correct**?

- A. G_2 – Synthesis phase
- B. S – Duplication phase
- C. G_1 – Chiasmata
- D. G_0 – Quiescent stage

Choose the most appropriate answer from the options given below:

- (1) One (2) Two
- (3) Three (4) Four

Q146 Select the **incorrect** statement w.r.t homologous chromosome.

- (1) They synapse during meiosis I.
- (2) They segregate from each other during meiosis I.
- (3) They come from only one of the individual's parents.
- (4) Genes present in them are same.

Q147 Given below are two statements:

Statement I: Viruses did not find a place in classification.

Statement II: Virus means venom or poisonous fluid.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct and Statement-II is incorrect
- (2) Statement-I is incorrect and Statement-II is correct
- (3) Both Statement-I and Statement-II are correct
- (4) Both Statement-I and Statement-II are incorrect

Q148 Select the **incorrect** statements.

- A. Classification is the process by which anything is grouped into convenient categories based on some easily observable characters.
- B. Based on characteristics, all living organisms can be classified into different taxa. This process of classification is taxonomy.
- C. Naming is only possible when the organism is described correctly and we know to what organism the name is attached to. This is nomenclature.

D. The scientific term for the categories is systematics.

Choose the most appropriate answer from the options given below:

- (1) A and C only (2) B and D only
- (3) C and D only (4) B and C only

Q149 Select the **incorrect** statements

A. The organisation of the prokaryotic cell is fundamentally similar even though prokaryotes exhibit a wide variety of shapes and functions.

B. The prokaryotic cells are represented by bacteria, blue-green algae, mycoplasma and PPLO.

C. All prokaryotes have a cell wall surrounding the cell membrane.

D. Prokaryotes are generally smaller and multiply more rapidly than the eukaryotic cells.

Choose the most appropriate answer from the options given below:

- (1) A and B only
- (2) A, B and D only
- (3) C only
- (4) A and D only

Q150 The common taxonomic categories of mango and wheat are:

- (1) class and division.
- (2) division and kingdom.
- (3) order and class.
- (4) order and family.

Q151

SECTION - A (ZOOLOGY)

Which of the following features is common among scorpion, silkworm, locust and honey bee?

- (1) Electric organ
- (2) Chitinous exoskeleton
- (3) Radula
- (4) Poison glands

Q152 Read the following statements (I-V).

I. Anterior part of genital pouch of female cockroach contains gonopore, spermathecal pores and collateral glands.



- II. Each compound eye of cockroach consists of about 2000 hexagonal ommatidia.
 III. The brain in cockroach is represented by sub-oesophageal ganglion.
 IV. 10 pairs of spiracles are present on the lateral surface of body in cockroach.
 V. Head is formed by the fusion of six segments and shows great mobility in all directions due to flexible neck.

Which of the following statements are **correct** w.r.t *Periplaneta americana*?

- (1) I, II and III only
 (2) III and IV only
 (3) I, II, IV and V only
 (4) I, II, III, IV and V

- Q153** All of the following statements are correct, **except:**
 (1) Anal cerci are present on the 10th segment of male and female cockroach.
 (2) In frogs, urinary bladder is present dorsal to the rectum.
 (3) Only adult cockroaches have wings.
 (4) In frogs, spinal cord is enclosed in vertebral column.

- Q154** The blood leaving the lungs has its oxygenated haemoglobin and gives up oxygen to the tissues because:
 (1) pO_2 in tissues is low and low H^+ concentration in tissues as compared to the lungs.
 (2) pO_2 in tissues is low and pCO_2 is high in tissues as compared to the lungs.
 (3) pO_2 in tissues is high and pCO_2 is low in tissues as compared to the lungs.
 (4) both (1) and (3)

- Q155** How much CO_2 delivered to alveoli by every 2 litre of deoxygenated blood approximately?
 (1) 4 mL (2) 4.2 mL
 (3) 8 mL (4) 80 mL

- Q156** Which of the following options is **correct** w.r.t uracil?
 (1) It is a nucleotide.
 (2) It possesses two nitrogen atoms.

- (3) It is a single ring structure.
 (4) Both (2) and (3)

- Q157** Choose the amino acid among the following that possesses an extra amino group in its structure.
 (1) Glycine (2) Lysine
 (3) Alanine (4) Serine

- Q158** The partial pressure of CO_2 in tissue:
 (1) is more than the pCO_2 of alveoli.
 (2) equals to the pCO_2 of oxygenated blood.
 (3) less than the pCO_2 of deoxygenated blood.
 (4) less than the pCO_2 of atmospheric air.

- Q159** Given below are two statements:
Statement I: Antibiotics and coloured pigments are secondary metabolites.
Statement II: Enzymes are denatured at low temperature as most enzymes are lipid in nature. In the light of the above statements, choose the most appropriate answer from the options given below:
 (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Statement I and Statement II both are correct.
 (4) Statement I and Statement II both are incorrect.

- Q160** The most abundant protein in the animal world is present in all of the following, **except:**
 (1) tendons.
 (2) ligaments.
 (3) cartilage.
 (4) blood.





Which of the following is **correct** w.r.t. the diagram shown above?

- (1) Presence of ventral mouth.
- (2) Presence of terminal mouth.
- (3) Presence of 6–15 pairs of gill slits for respiration.
- (4) Presence of forwardly directed scales.

Q162 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): The polysaccharide which is present mainly as the store house of energy in plant tissues can hold I_2 molecules.

Reason (R): Main storage polysaccharide of plants does not contain complex helices that can hold I_2 molecules.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q163 A student wanted to study about gap junctions in more detail. He can do so by analysing the muscle fibres which are characterised by:

- I. Present in heart
- II. Striations with intercalated discs
- III. Branching pattern

Choose the **correct** option.

- (1) I and III only

- (2) I and II only
- (3) II and III only
- (4) I, II and III

Q164 Choose the **correct** option w.r.t. secondary metabolites.

- (1) Alkaloid – Concanavalin A
- (2) Drug – Carotenoids
- (3) Toxin – Abrin
- (4) Lectin – Curcumin

Q165 Which of the following organisms has a unique feature of having hairs on skin?

- (1) *Neophron*
- (2) *Rattus*
- (3) *Calotes*
- (4) *Alligator*

Q166 Arrange the following in the order of increasing volume and choose the **correct** option.

- I. Tidal volume
- II. Expiratory reserve volume
- III. Inspiratory reserve volume
- IV. Vital capacity

- (1) $I < II < III < IV$
- (2) $I < III < II < IV$
- (3) $I < IV < III < II$
- (4) $I < IV < II < III$

Q167 Bone differs from cartilage in possessing:

- (1) solid and pliable ground substance.
- (2) chondrocytes are present in the fluid filled spaces called lacunae.
- (3) matrix which is rich in calcium salts and collagen fibres.
- (4) intercellular material that resists compression.

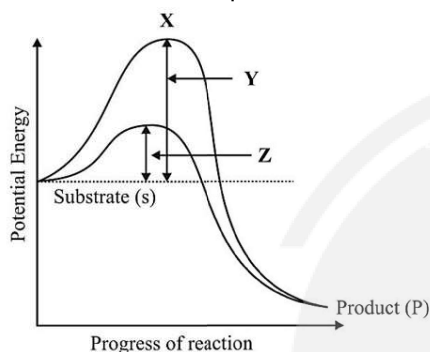
Q168 Choose the **incorrect** statement.

- (1) Hydrogen bond is least likely to be involved in stabilising 3-D structure of most proteins.
- (2) In proteins, only right-handed helices are observed.
- (3) Adult human haemoglobin consists of two subunits of α -type and two subunits of β -type.
- (4) Tertiary structure is absolutely necessary for the many biological activities of proteins.



- Q169** The type of cells that make up more than one half the volume of neural tissues in our body:
- (1) are specialised to transmit impulses in our body.
 - (2) exert the greatest control over the body's responsiveness to changing conditions by transmitting impulses.
 - (3) are excitable cells due to the presence of a differential concentration gradient of ions.
 - (4) protect and support the cells that act as the functional unit of neural system.

- Q170** Consider the given below labelled graph and choose the **correct** option.



- (1) The reaction shown in the graph represents an endothermic reaction.
 - (2) 'X' is transient and unstable.
 - (3) 'Y' represents the activation energy in the presence of enzyme.
 - (4) 'Z' represents the activation energy without enzyme.
- Q171** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
- Assertion A:** *Felis* are homeotherms.
- Reason R:** *Felis* can maintain a constant body temperature.
- In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is not the correct explanation of A.

- Q172** The enzyme that catalyses the hydrolysis of bonds present between the monomeric units of chitin, belongs to which class of enzymes?
- (1) Dehydrogenase
 - (2) Transferase
 - (3) Hydrolase
 - (4) Ligase

- Q173** Match **List-I** with **List-II**:

List-I		List-II	
(A)	Squamous epithelium	(I)	Assists in secretion and absorption in the proximal convoluted tubule of kidney.
(B)	Cuboidal epithelium	(II)	Provides protection against chemical stress.
(C)	Ciliated epithelium	(III)	Forms diffusion boundary.
(D)	Compound epithelium	(IV)	To move mucus in a specific direction.

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
 - (2) A-II, B-I, C-III, D-IV
 - (3) A-IV, B-III, C-I, D-II
 - (4) A-I, B-IV, C-II, D-III
- Q174** GLUT-4 belongs to the similar category of biomolecules to which 'X' also belongs. Identify 'X' and choose the **correct** option.
- (1) Inulin
 - (2) Glycogen
 - (3) Antibody
 - (4) Chitin
- Q175** Choose the **correct** function w.r.t the most abundant and widely distributed tissue in the body of complex animals.
- (1) Linking and supporting other tissues/organs of the body.
 - (2) Lines body cavities, ducts and tubes.



- (3) Plays an active role in all the movements of the body.
 (4) Causes stimulation and inhibition of adjacent neurons and other cells.

Q176 Read the following statements (I-IV).

- I. CO_2 is carried by haemoglobin as carbamino-haemoglobin and this binding is related to the partial pressure of CO_2 .
 II. Respiration is regulated by the large numbers of chemoreceptors located in the carotid bodies and in the aortic bodies.
 III. The trachea, primary, secondary and tertiary bronchi and initial bronchioles are supported by complete cartilaginous rings.
 IV. Emphysema is caused due to damage in alveolar walls and smoking is one of the major causes of it.

Which of the following statements are **correct**?

- (1) I, II and III only
 (2) III and IV only
 (3) I, II and IV only
 (4) I, II, III and IV

Q177 The common feature between the muscles that are present in the wall of intestine and in the biceps is '**X**', while the common feature between the muscles that are present in heart and in the limbs is '**Y**'.

Choose the option that **correctly** identifies '**X**' and '**Y**' respectively.

- (1) X-Presence of striated muscle fibres, Y-Involuntary in action
 (2) X-Presence of unbranched fibres, Y-Presence of muscle fibres with central nucleus
 (3) X-Presence of fusiform shaped muscle fibres, Y-Presence of striations
 (4) X-Absence of intercalated discs, Y-Cylindrical shaped muscle fibres

Q178 Certain organic compounds form 3% of the total cellular mass of the living tissue. In polymeric form of such organic compounds, monomeric units are joined together with the help of **X**.

X is also present in all of the following, **except**:

- (1) cotton fibre. (2) paper pulp.

- (3) RuBisCO. (4) glycogen.

Q179 Given below are two statements:

Statement I: Sulphate and phosphate are present in the acid soluble fraction of living tissue.

Statement II: Arachidonic acid has 16 carbon atoms including the carboxyl carbon.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Statement I and Statement II both are correct.
 (4) Statement I and Statement II both are incorrect.

Q180 Hepatic caeca in cockroaches are:

- (1) 6-8 in number, a part of excretory system and is present at the junction of midgut and hindgut.
 (2) 9-12 in number, a part of the digestive system and is present at the junction of midgut and hindgut.
 (3) 6-8 in number, a part of the digestive system and is present at the junction of foregut and midgut.
 (4) 9-12 in number, a part of excretory system and is present at the junction of foregut and midgut.

Q181 Choose the **odd** one out w.r.t radial symmetry.

- (1) *Pleurobrachia* (2) *Physalia*
 (3) *Taenia* (4) Adult *Echinus*

Q182 Which of the following factors is responsible for the dissociation of oxyhaemoglobin?

- (1) High pO_2
 (2) Low temperature
 (3) Low H^+
 (4) High pCO_2



Q183 Match the **List-I** with **List-II**:

List-I		List-II	
(A)	Mammary glands	(I)	<i>Saccoglossus</i>
(B)	Proboscis gland	(II)	<i>Obelia</i>
(C)	Alternation of generation	(III)	Platypus
(D)	Bioluminescence	(IV)	<i>Ctenoplane</i>

Choose the **correct** answer from the options given below:

- (1) A- III, B-I, C-IV, D-II
- (2) A-IV, B-II, C-III, D-I
- (3) A-III, B-I, C-II, D-IV
- (4) A-I, B-III, C-IV, D-II

Q184 The prosthetic group of the enzyme 'peroxidase' also acts as the co-factor for the enzyme named:

- (1) Carboxypeptidase.
- (2) Catalase.
- (3) Carbonic anhydrase.
- (4) Succinic dehydrogenase.

Q185 Select the **correct** pair of primary metabolites whose monomeric unit is glucose.

- (1) Chitin and glycogen
- (2) Glycogen and starch
- (3) Glucagon and inulin
- (4) Sucrose and insulin

Q186 **SECTION - B (ZOOLOGY)**

The total number of alkyl groups present in the triglyceride is equal to the total number of:

- (1) hydroxyl groups in a glycerol.
- (2) amino acids in a lecithin.
- (3) hydrogen atoms directly attached to the alpha C atom in glycine.
- (4) hydrogen bonds present between adenine and thymine.

Q187 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Pleural fluid is present between double layered pleural membrane of the lungs.

Reason R: Pleural fluid enhance the friction between the pleural membrane.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q188 Which of the following is a **correct** matching pair of body features and the animal possessing it?

- (1) Vertebral column - *Salpa*, *Petromyzon*
- (2) Gnathostomata - *Corvus*, *Myxine*
- (3) Air bladder - *Labeo*, *Pristis*
- (4) Pneumatic bones - *Corvus*, *Columba*

Q189 Given below are two statements:

Statement I: Trachea is a straight tube which divides at the level of 3rd thoracic vertebra.

Statement II: Solubility of oxygen is 20-25 lower than the carbon dioxide.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Statement I and Statement II both are correct.
- (4) Statement I and Statement II both are incorrect.

Q190 Choose the **correct** match.

(1)	Tendons	–	Attach one bone to another bone
(2)	Ligaments	–	Attach skeletal muscles to bones
(3)	Ducts of glands	–	Composed of a layer of cube-like cells
(4)	Neurons	–	Possess axon and neuroglia

Q191 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: In inspiration, diaphragm becomes flat shaped.



Reason R: In inspiration, contraction of diaphragm muscle decreases the volume of thoracic cavity.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

Q192 Refer to the following structures.

Eyes, Antennae, Labial palps, Maxillary palps, Anal cerci

How many of the following act as sensory organs in cockroach?

- (1) Four
- (2) Three
- (3) Five
- (4) Two

Q193 Read the following statements (A-D).

- A. The body cavity which is lined by mesoderm is present in *Nereis*.
- B. Notochord is a rod-like structure formed on the dorsal side during embryonic development in *Betta*.
- C. *Euspongia* exhibit cellular level of organisation.
- D. *Neophron* and *Macaca* are warm blooded animals.

Choose the option with the **correct** statements:

- (1) A and B only
- (2) A and C only
- (3) A and D only
- (4) A, B, C and D

Q194 Inhibition of succinic dehydrogenase by malonate can be reversed by increasing the:

- (1) concentration of enzyme.
- (2) concentration of substrate.
- (3) pH of the reaction.
- (4) temperature of the reaction.

Q195 Which of the following features distinguishes a male frog from a female frog?

- (1) Absence of copulatory pads on the forelimbs.
- (2) Presence of copulatory pads on the hind wings.
- (3) Absence of sound producing vocal sacs.
- (4) Presence of sound producing vocal sacs.

Q196 Match List-I with List-II:

List-I		List-II	
(A)	Chitin	(I)	Contains vitamin niacin
(B)	Collagen	(II)	Exoskeleton of arthropods
(C)	Anthocyanins	(III)	Intercellular ground substance
(D)	Coenzyme NADP	(IV)	Secondary metabolite belonging to the category of pigments

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-II, B-IV, C-III, D-I
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-III, C-IV, D-I

Q197 Choose the **incorrect** statement.

- (1) Amphibians, reptiles, birds and mammals respire through lungs only.
- (2) The thoracic chamber is formed dorsally by the vertebral column, ventrally by the sternum, laterally by the ribs and on the lower side by the dome-shaped diaphragm.
- (3) Total volume of air accommodated in the lungs at the end of a forced inspiration is total lung capacity.
- (4) O₂ can bind with haemoglobin in a reversible manner to form oxyhaemoglobin.

Q198 Read the following statements (I-V).

- I. *Aptenodytes* and *Chameleon* are oviparous.
- II. In *Doliolum*, notochord extends from head to tail region.
- III. *Myxine* are freshwater organisms but they migrate to marine water for spawning.



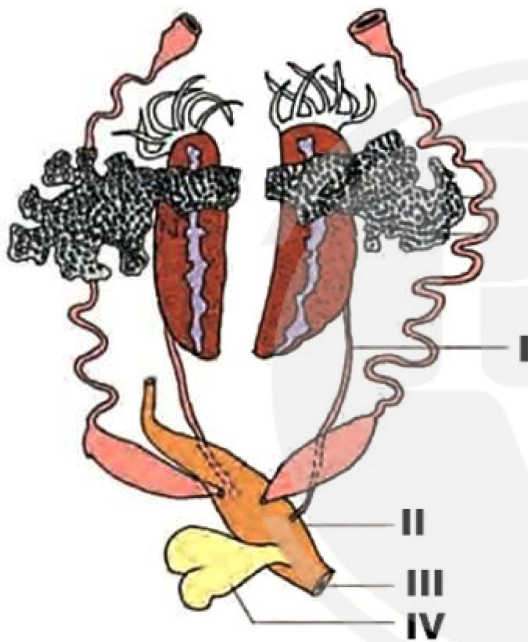
IV. The mouth of *Carcharodon* is located ventrally.

V. *Macropus* and *Balaenoptera* show direct development.

Which of the following statements are **incorrect**?

- (1) I, II and III only
- (2) III and IV only
- (3) II and III only
- (4) I, II, III and V only

Q199 The given diagram is of female reproductive system of frogs. Identify the structure responsible for storage of urine before it is excreted out.



- (1) I
- (2) II
- (3) III
- (4) IV

Q200 Match **List-I** with **List-II**:

List-I (Animal)		List-II (Respiratory Organ)	
A.	<i>Scoliodon</i>	I.	Tracheal tubes
B.	<i>Delphinus</i>	II.	Lungs
C.	<i>Hydra</i>	III.	Entire body surface
D.	Insects	IV.	Gills

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-I, B-III, C-IV, D-II
- (4) A-IV, B-II, C-III, D-I



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